

Solution Tasks

Task 1: Capstone Project Selection & Justification

- **Selected Capstone Option:** Healthcare Management API (Cardiac Patient Monitoring System)
- **Technical Justification:**

"This project aligns with my core technical competencies in normalized relational database design (3NF), asynchronous programming patterns in C#, and the implementation of secure REST API architectures. The healthcare domain provides a highly practical environment to apply critical software engineering concepts—such as role-based access control (RBAC), data-ownership verification, and strict input validation—which are essential for protecting sensitive clinical data."

Task 2: 3-Sentence Scope Statement (Professional Baseline Confirmation)

(Database & Core Architecture): The Cardiac Patient Monitoring System delivers an asynchronous REST API backed by a normalized SQL Server database configured via Entity Framework Core code-first migrations and explicit Fluent API relationships. **(Security & Hardening):** The application enforces robust security and data integrity by integrating ASP.NET Core Identity, JWT bearer authentication with token rotation, role-based and ownership-based access control, centralized exception middleware, and strict request validation through FluentValidation. **(Quality Assurance & Delivery):** By Week 9, the project will meet the required professional baseline with automated unit and integration testing (xUnit, Moq, and WebApplicationFactory), comprehensive Swagger and Postman documentation, and a deployed continuous integration and delivery pipeline (CI/CD) on a cloud hosting platform.